

Student Guide — primary / module-14- magic-squares

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Magic Squares — Bhadragaṇita (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *bhadra-gaṇita* — _____
- *siddhāṅka* (modern coinage) — _____
- *kothumi vidhi* — _____

Day-by-day notes

Day 1 — Discover the 3×3

What I learned:

One question I have:

Day 2 — The Magic Constant

What I learned:

One question I have:

Day 3 — How Many 3×3 Magic Squares?

What I learned:

One question I have:

Day 4 — Stair-Step Method (Nārāyaṇa)

What I learned:

One question I have:

Day 5 — Practice — 5×5 and 7×7

What I learned:

One question I have:

Day 6 — 4×4 and the Khajuraho Square

What I learned:

One question I have:

Day 7 — Historical Context — Nārāyaṇa and Predecessors

What I learned:

One question I have:

Day 8 — Cross-Cultural — Lo Shu and Dürer

What I learned:

One question I have:

Day 9 — Design Your Own

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Discover the 3×3

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Build the unique 3×3 magic square with 1–9 by trial.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. "auspicious arithmetic" — the Indian term for magic squares)
- BIG puzzle! Put the numbers 1 to 9 in a 3×3 grid. Each row, column, and diagonal should add to the SAME number!

Children work in pairs. Try, fail, try again.

Hint after 5 min: *"5 always goes in the middle."*

The answer: 2-7-6 / 9-5-1 / 4-3-8. All rows = 15! All columns = 15! Even diagonals = 15! MAGIC!

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Magic Constant

Module: Magic Squares — Bhadrageṇita (Primary) · **Time:** 30 min

Goal

Derive why 3×3 with 1–9 must sum to 15 per row.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)
- The MAGIC NUMBER for 3×3 is 15. Why 15?

Add all the numbers from 1 to 9: $1+2+3+4+5+6+7+8+9 = 45$.

Spread 45 across 3 rows: $45 / 3 = 15$. So each row must equal 15!

For a 4×4 magic square (using 1 to 16), what's the magic number? - $1+2+\dots+16 = 136$. / 4 rows = **34**.

“*There's a pattern!*”

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 3 — How Many 3×3 Magic Squares?

Module: Magic Squares — Bhadrageṇita (Primary) · **Time:** 30 min

Goal

Count the equivalence classes (up to symmetry).

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa’s algorithm for odd-order squares”)
- How many ways can we make a 3×3 magic square?

Show the 2-7-6 / 9-5-1 / 4-3-8 square. Rotate the paper 90° — STILL magic! Rotate again — still! Reflect (mirror) — still!

“That’s 8 ways! But they’re all REALLY the same square — just turned around.”

Children try rotating their own magic square paper. They count the rotations.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 4 — Stair-Step Method (Nārāyaṇa)

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Apply Nārāyaṇa’s algorithm to build a 5×5 square.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: *“What’s one thing you remember from yesterday?”*

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

- Big-kid trick! Nārāyaṇa (an Indian mathematician 700 years ago) taught how to BUILD a magic square step-by-step.

Show on the board: 1. Put **1** in the top-middle. 2. From 1, jump UP-and-RIGHT to put 2... but you fall off the top! 3. Wrap to the bottom row (same column). 4. Keep going!

The pattern looks like a horse jumping. “*That’s why he called it the horse-step!*”

Children watch the teacher build a 3×3 step-by-step.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Practice — 5×5 and 7×7

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Use the method on multiple odd-order squares.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)
- Today: try a BIGGER magic square — 5×5 !

Place 1 in the top-middle. Step up-right. Wrap when needed. Keep going to 25.

It looks confusing at first, but the rule is the same. The magic number for 5×5 is **65** (all rows, columns, diagonals add to 65).

Children try with the teacher's help. Most won't finish — that's OK! Just enjoy the start.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — 4×4 and the Khajuraho Square

Module: Magic Squares — Bhadraganita (Primary) · **Time:** 30 min

Goal

Examine the Khajuraho inscription and its panel-magic property.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)
- 4×4 magic squares are also amazing! In INDIA, there's one carved on a temple wall — the Khajuraho square. About 1000 years old!

Show the picture. “*All rows = 34, all columns = 34. Even small 2x2 boxes inside = 34!*”

Children draw a 4×4 grid and copy the numbers. They check: do the rows add to 34? Try one!

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Place *Gaṇita-kaumudī* in 14th c. Indian math.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. "auspicious arithmetic" — the Indian term for magic squares)
- Nārāyaṇa Paṇḍita lived in India about 700 years ago. He wrote a book about magic squares!

His book is called *Gaṇita-kaumudī* ("Moonlight of Math"). Book 14 is all about magic squares.

Show a (drawn) picture of Nārāyaṇa with a manuscript. "Maths heroes from long ago!"

Children draw a magic square inside a "book" outline.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Cross-Cultural — Lo Shu and Dürer

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. "the common sum of all rows/columns/diagonals")
- Magic squares are found ALL OVER THE WORLD!
-  China: Lo Shu legend — a turtle's shell!
-  India: Khajuraho temple wall!
-  Germany: Dürer's painting!

Show 4 small pictures. "People everywhere love magic squares!"

Each child picks one tradition and draws a tiny picture (turtle, temple, painting).

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Design Your Own

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Create a personalised 4×4 or 5×5 magic square as visual art.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)
- Make YOUR OWN magic square as art!

Choose ANY 9 numbers (or use 1-9). Make a 3×3 grid where all rows add to the same. Decorate it!

Show example: a magic square shaped like a flower, with colours on each number. “*Math + art = magic!*”

Children draw, colour, and write their magic number at the top.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Share and Assessment

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Present design + summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)
- Final day! Show your magic square art.

Each child shows + says one thing they liked. Cheer for each.

Quiz with stickers — everyone gets a “Magic Squares” sticker.

End: chant — “*Bhadragaṇita! Magic numbers!*”

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — Magic Squares — Bhadragaṇita (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Discover the 3×3** · Draw about today's concept. (~5 min)
2. **Day 2 — The Magic Constant** · Draw about today's concept. (~5 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Draw about today's concept. (~5 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Draw about today's concept. (~5 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Draw about today's concept. (~5 min)
6. **Day 6 — 4×4 and the Khajuraho Square** · Draw about today's concept. (~5 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Draw about today's concept. (~5 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Draw about today's concept. (~5 min)
9. **Day 9 — Design Your Own** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.